Ericsson PE (WFD-PE-02) Optical Transport Failure Standard Operating Procedure (ALARM_2792)

Node: WFD-PE-02 | **Vendor:** Ericsson | **Severity:** Major | **Alarm Code:** ALARM_2792

Description

The Optical Transport Failure in the Ericsson PE (WFD-PE-02) at the Whitefield POP is a critical failure domain that occurs due to the degradation of SFP/XFP modules, fiber cuts, or Rx/Tx power loss. This failure is specific to the PE role and Ericsson equipment due to the underlying architecture of the optical transport system. The PE node is responsible for terminating the optical transport connections and forwarding data packets to the core network. The SFP/XFP modules are crucial components that enable the PE node to transmit and receive optical signals. When these modules degrade or experience power loss, it can lead to signal degradation, errors, or complete loss of connectivity. The fiber cuts or damage can also cause signal degradation or loss of connectivity. In such cases, the PE node will raise an alarm (ALARM_2792) to indicate a major failure. To resolve this failure, it is essential to identify the root cause, isolate the issue, and execute the necessary steps to repair or replace the faulty components.

Pre-requisite Checks

- Verify that the redundant link is working properly.
- Ensure that the remote hands are dispatched to the Whitefield POP for physical access.
- Check the environmental conditions at the site, including temperature and humidity levels.
- Verify that the power supply to the PE node is stable and within the recommended range.

Potential Root Causes

Cause	Probability
Degradation of SFP/XFP modules	High
Fiber cuts or damage	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show opt trans port status, Expected Output: Optical transport port status: [insert status here], Description: This command is used to display the status of the optical transport ports. The output will indicate the current status of the ports, including any errors or alarms.

Command: show opt trans port detailed, Expected Output: Optical transport port detailed status: [insert status here], Description: This command is used to display detailed information about the optical
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transport ports, including the SFP/XFP module status, Rx/Tx power
levels, and fiber status.

Command: show opt trans port stats, Expected Output: Optical transport
port statistics: [insert statistics here], Description: This command
is used to display statistical information about the optical transport
ports, including error counts, packet counts, and signal quality
metrics.
```

Resolution Steps

Step 1: Step: Isolate the faulty SFP/XFP module, Command: replace opt trans port [insert port number] with [insert new SFP/XFP module], Description: This step involves replacing the faulty SFP/XFP module with a new one. The new module should be installed in the same port as the faulty one.

Step 2: Step: Verify the fiber connectivity, Command: show opt trans port detailed, Description: This step involves verifying the fiber connectivity to ensure that there are no issues with the fiber cable or connectors.

Step 3: Step: Adjust the Rx/Tx power levels, Command: set opt trans port [insert port number] rx power [insert value] and tx power [insert value], Description: This step involves adjusting the Rx/Tx power levels to optimize the signal quality. The values should be adjusted according to the recommended settings for the specific SFP/XFP module.

Post-Check Verification

```
Command: show opt trans port status, Expected Output: Optical
transport port status: [insert status here], Description: This command
is used to verify that the optical transport ports are functioning
correctly.

Command: show opt trans port detailed, Expected Output: Optical
transport port detailed status: [insert status here], Description:
This command is used to verify that the SFP/XFP modules, Rx/Tx power
levels, and fiber status are functioning correctly.

Command: show opt trans port stats, Expected Output: Optical transport
port statistics: [insert statistics here], Description: This command
is used to verify that the statistical information about the optical
transport ports is accurate.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: replace opt trans port [insert port number] with [insert original SFP/XFP module], Description: This step involves restoring the original SFP/XFP module that was replaced during the resolution process.

- Step: Restore the original fiber connectivity, Command: show opt trans port detailed, Description: This step involves restoring the original fiber connectivity that was verified during the resolution process.

- Step: Restore the original Rx/Tx power levels, Command: set opt trans port [insert port number] rx power [insert original value] and tx power [insert original value], Description: This step involves restoring the original Rx/Tx power levels that were adjusted during the resolution process.